How-To Library

Informatica

Processing Australia Addresses in Certified Mode

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Abstract

Informatica Address Verification can verify addresses to the Address Matching Approval System (AMAS) standards of Australia Post. This article describes the values that Address Verification can return for Australia addresses in certified mode. The article also answers some common questions about address verification to the AMAS standard.

Supported Versions

- Informatica Address Verification 3.0 and later
- Informatica Address Verification (On-Premises) 5.14.0

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Introduction

You can use the certified process mode in Informatica Address Verification to verify Australia addresses to the AMAS standards. Address Verification is approved at this time to certify addresses in accordance with AMAS Cycle 2019.

To verify Australia addresses in certified mode, install the `AUS5C1.MD` and `AUS5C2.MD` databases and add the corresponding unlock codes to `SetConfig.xml`. The `AUS5C1.MD` and `AUS5C2.MD` databases contain Postal Address File (PAF) data, which includes Australia Post's Delivery Point Identifiers (DPIDs).

Attributes on the AMAS Element for Australia Addresses

The AMAS standard requires that each address in an address set includes values that can affirm the deliverability and accuracy of the address. Informatica Address Verification can return the values when you process the addresses in certified mode.

Address Verification returns the values as `Type` attributes on the *AMAS* element. Find the AMAS element under the `EnrichmentData` element in the `Result.xml` file.

The following table describes the values:

Field	Attribute ID	Description
Error Code	ERRORCODE	A two-character code that represents the validity of the address with respect to the AMAS standard. The code can indicate that the address is fully valid, or the code can describe the reason why the address does not meet the AMAS standard. For more information on error codes, see "AMAS Error Codes in Australia Addresses" on page 4.
Delivery Point Identifier	DELIVERY_POINT_ID	An eight-digit identifier that Australia Post assigns to a delivery point. Each DPID is unique and randomly generated. Australia Post creates DPIDs in a range from 30,000,000 to 99,999,999.
Lot Number	LOT_NBR	The lot reference number that a government department assigned to a property during the sub-division of a parcel of land and prior to road numbering.
Postal Delivery Number	POSTAL_DELIVERY_NBR	The number information in the delivery service. For example, the post office box number in a P.O. Box address. Australia Post stores a postal delivery number as a five-digit number in the Postal Address File (PAF). Therefore, the PAF stores the postal delivery number 500 as 00500. If the address does not contain a postal delivery number, Address Verification does not populate the attribute.
Postal Delivery Number Prefix	POSTAL_DELIVERY_NBR_PFX	The prefix character in a postal delivery number. For example, A in A500.
Postal Delivery Number Suffix	POSTAL_DELIVERY_NBR_SFX	The suffix character in a postal delivery number. For example, A in 500A.
Postal Delivery Type	DESCRIPTOR	The type of postal delivery service. **Note:** Address Verification does not return the postal delivery type under the AMAS element. Find the DESCRIPTOR attribute for the postal delivery type on the *DeliveryService* element under the *AddressElements* element in the `Result.xml` file. For more information on the delivery type, see "Postal Delivery Types in Australia Addresses" on page 4.
House Number 1	HOUSE_NBR_1	The primary house number in a street address. Australia Post stores a house number as a five-digit number in the PAF. Therefore, the PAF stores the house number 123 as 00123.
House Number 1 Suffix	HOUSE_NBR_SFX_1	An alphabetical suffix to a House Number 1 value. For example, A in 123A.
House Number 2	HOUSE_NBR_2	The secondary house number in a street address. Australia Post stores a house number as a five-digit number in the PAF. Therefore, the PAF stores the house number 456 as 00456. If the address does not contain a secondary house number, Address Verification returns 00000 as the attribute value.

Field	Attribute ID	Description
House Number 2 Suffix	HOUSE_NBR_SFX_2	An alphabetical suffix to a House Number 2 value. For example, B in 456B.
Record Type	RECORD_TYPE	An alphabetic value that identifies the type of address. For more information on record types, see "Record Type Indicators for Australia Addresses " on page 5.

Postal Delivery Types in Australia Addresses

Address Verification returns the postal delivery type for Australia addresses on the `Type` attribute of the `DeliveryService` element. To read the postal delivery type, set the `Type` attribute on the *DeliveryService* element to DESCRIPTOR.

Address Verification can return the following types:

- CARE PO. Care of post office
- CMA. Community mail agent
- CMB. Community mail bag
- GPO BOX. General post office box
- LOCKED BAG. Locked mail bag service
- MS. Mail service
- PO BOX. Post office box
- CARE PO. Poste restante
- PRIVATE BAG. Private mail bag service
- RSD. Roadside delivery
- RMB. Roadside mail bag
- RMB. Roadside mail box
- RMS. Roadside mail service
- CPA. Community postal agent

AMAS Error Codes in Australia Addresses

Address Verification returns the following AMAS error codes:

- E0. The address is valid to the AMAS standard. No changes are necessary.
- E1. The address is not valid to the AMAS standard. To include the record in the record set you submit for AMAS certification, delete all address data.
- E2. The address is not valid to the AMAS standard. One or more issues exist in street data.

  To include the record in the record set you submit for AMAS certification, delete all street data and house number data.
- E3. The address is not valid to the AMAS standard. One or more issues exist in the delivery service data.

  To include the record in the record set you submit for AMAS certification, delete all delivery service or Post Office Box data.

- E4. The address is not valid to the AMAS standard. One or more issues exist in the street name data.

  To include the record in the record set you submit for AMAS certification, delete the street name.
- E5. The address is not valid to the AMAS standard. One or more issues exist in the house number data.

  To include the record in the record set you submit for AMAS certification, delete the house number.
- E6. The address is not valid to the AMAS standard. One or more issues exist in the delivery service number.

  To include the record in the record set you submit for AMAS certification, delete the delivery service or Post Office Box number.
- EX. Address validation cannot process the address to the AMAS standard. The address format is not valid.

Record Type Indicators for Australia Addresses

Informatica Address Verification returns the following address type values for Australia addresses:

- B. The address identifies a building.
- F. The address identifies an organization.
- L. The address post code identifies the organization as a large volume receiver. The reference data adds or validates the organization name.
  Address Verification can determine that the address is a large volume receiver in one of the following ways:
  - The address post code identifies the organization as a large volume receiver.
  - The reference data does not contain street or building information.
- P. The address identifies a Post Office Box or a delivery service.
- S. The address identifies a street. S is the default address type. If Address Verification cannot determine the address type from the address data, it returns the default value.
- U. Unidentified. The address is not valid, and Address Verification does not assign an address type.

If an address meets the criteria for more than one address type, Address Verification assigns the first applicable address type from the following list:

L, F, P, B, S

Verifying an Address in Certified Mode

Address Verification returns the AMAS response in the result output when you verify an Australia address in certified mode.

Input and Result XML in Address Verification (On-Premises)

The following example shows a request to validate an Australia address in certified mode.

Note the following parameter values in the input XML:

- The `ProcessMode` value is `CERTIFIED`.
- The `AMAS` attribute of the `Process` element is `ON`.

You enter the following input address:

```
<InputData>
    <AddressElements>
        <Country Item="1" Type="NAME">AUS</Country>
    </AddressElements>
    <AddressLines>
        <FormattedAddressLine Line="1">SHOP 7 208 ADELAIDE ST</FormattedAddressLine>
        <FormattedAddressLine Line="2">BRISBANE CITY QLD 4000</FormattedAddressLine>
```

```
    </AddressLines>
</InputData>
```

Informatica Address Verification (On-Premises) returns the following result, which includes enrichments available in the certified mode:

```
<?xml version="1.0" encoding="UTF-16"?>

<!-- EngineVersion="5.13.1.45876" -->

<Result     ProcessStatus="V4"
     ModeUsed="CERTIFIED"
     Count="1"
     CountOverflow="NO"
     CountryISO3="AUS"
     PreferredScript="LATIN_1"
     PreferredLanguage="ENGLISH">

<ResultData ResultNumber="1"
     MailabilityScore="5"
     ResultPercentage="100.00"
     ElementResultStatus="F0F0F0F0F00000F000E0"
     ElementInputStatus="60606060600000600060"
     AddressResolutionCode="00000000000000000000"
     ExtElementStatus="00000000000000000000"
     ElementRelevance="10101010100000100010"
     AddressType="S"
     LanguageISO3="">
    <AddressElements>
        <Country Type="NAME_EN" Item="1">Australia</Country>
        <Locality Item="1">Brisbane City</Locality>
        <PostalCode Item="1">4000</PostalCode>
        <Province Item="1">QLD</Province>
        <Street Item="1">Adelaide St</Street>
        <Number Item="1">208</Number>
        <SubBuilding Item="1">Shop 7</SubBuilding>
    </AddressElements>
    <AddressLines>
        <DeliveryAddressLine Line="1">Shop 7 208 Adelaide St</DeliveryAddressLine>
        <CountrySpecificLocalityLine Line="1">Brisbane City  QLD  4000</
CountrySpecificLocalityLine>
        <FormattedAddressLine Line="1">Shop 7 208 Adelaide St</FormattedAddressLine>
        <FormattedAddressLine Line="2">Brisbane City  QLD  4000</FormattedAddressLine>
        <FormattedAddressLine Line="3">Australia</FormattedAddressLine>
    </AddressLines>
    <AddressComplete>Shop 7 208 Adelaide St
Brisbane City  QLD  4000
Australia</AddressComplete>
    <EnrichmentData
         GeoCodingStatus="EGCN"
         CAMEOStatus="ECON"
         CASSStatus="ECA0"
         SERPStatus="ESE0"
         SNAStatus="ESN0"
         AMASStatus="EAM1"
         SendRightStatus="ESR0"
         SupplementaryGBStatus="EGB0"
         SupplementaryJPStatus="EJP0"
         SupplementaryRSStatus="ERS0"
         SupplementaryBRStatus="EBR0"
         SupplementaryDEStatus="EDE0"
         SupplementaryZAStatus="EZA0"
         SupplementaryCHStatus="ECH0"
         SupplementaryATStatus="EAT0"
         SupplementaryPLStatus="EPL0"
         SupplementaryFRStatus="EFR0"
         SupplementaryBEStatus="EBE0"
         SupplementaryRUStatus="ERU0"
         SupplementaryUSStatus="EUS0"
         SupplementaryKRStatus="EKR0"
         SupplementaryITStatus="EIT0"
```

```
            SupplementaryESStatus="EES0"
            SupplementaryCZStatus="ECZ0">
               <GeoCoding Type="COMPLETE"></GeoCoding>
               <GeoCoding Type="LATITUDE"></GeoCoding>
               <GeoCoding Type="LONGITUDE"></GeoCoding>
               <GeoCoding Type="LAT_LONG_UNIT"></GeoCoding>
               <CAMEO Type="CATEGORY"></CAMEO>
               <CAMEO Type="INTERNATIONAL"></CAMEO>
               <CAMEO Type="GROUP"></CAMEO>
               <CAMEO Type="MVID"></CAMEO>
               <CAMEO Type="CATEGORY_DESCRIPTION"></CAMEO>
               <CAMEO Type="GROUP_DESCRIPTION"></CAMEO>
               <CAMEO Type="INTERNATIONAL_DESCRIPTION"></CAMEO>
               <AMAS Type="ERRORCODE">E0</AMAS>
               <AMAS Type="RECORD_TYPE">S</AMAS>
               <AMAS Type="DELIVERY_POINT_ID">44686342</AMAS>
               <AMAS Type="LOT_NBR"></AMAS>
               <AMAS Type="POSTAL_DELIVERY_NBR">00000</AMAS>
               <AMAS Type="POSTAL_DELIVERY_NBR_PFX"></AMAS>
               <AMAS Type="POSTAL_DELIVERY_NBR_SFX"></AMAS>
               <AMAS Type="HOUSE_NBR_1">00208</AMAS>
               <AMAS Type="HOUSE_NBR_SFX_1"></AMAS>
               <AMAS Type="HOUSE_NBR_2">00000</AMAS>
               <AMAS Type="HOUSE_NBR_SFX_2"></AMAS>
        </EnrichmentData>
    </ResultData>
    </Result>
```

Frequently Asked Questions

What is the difference between a DPID and a DID?

A DIPD is an eight-digit identifier that Australia Post assigns to a delivery point. Each DPID is unique and randomly generated. Australia Post creates DPIDs in a range from 30,000,000 to 99,999,999.

A Delivery Identifier (DID) is an eight-digit number that Australia Post assigns to a Group or Locality record and that represents all of the addresses in the Group or Locality.

Australia Post creates Group DIDs in a range from 26,000,000 to 27,999,999. Australia Post creates Locality DIDs in a range from 28,000,000 to 28,999,999.

In some cases, AMAS certification is possible for addresses that return a DID and not a DPID. For example, Address Verification may identify a certified match on a record with a Group DID value and not a DPID value when it can find an exact match with a group record in the PAF.

What are the criteria for address certification to the AMAS standard?

Australia Post maintains a set of stringent criteria for the certification of any address to the AMAS standard. Australia Post publishes a handbook that describes the criteria. For complete information on the certification criteria, consult the AMAS handbook.

How does batch mode differ from certified mode?

Address verification in certified mode applies more strict standards for accuracy and completeness to an input address. Certified mode permits fewer corrections to an input address in the reference data matching stage. For example, if an input address fails to match an address in the PAF completely but can match an address if a house number suffix is added, certification does not permit the addition of the suffix and will not return an E0 code for the address.

The AMAS criteria can cause Address Verification to reject an address in certified mode that could produce a reference data match in batch mode. For example, the AMAS criteria state that an address verification process must reject a phantom primary point address.

A phantom primary point is an address that identifies a verifiable street destination but not a mailbox for delivery. That is, the address provides primary information, or information to street and building number level, but it does not provide required secondary information, such as a sub-building number. The street address of an apartment building is a phantom primary point if the building itself does not receive mail.

Note: You can add the Address Resolution Code field to the output address to identify missing elements in the address. An Address Resolution Code value of 2 in any position in the address structure indicates that a data element is absent but required at that position.

Why does Address Verification reject an address in certified mode when the street information falls on the border of two localities?

AMAS certification defines Bordering Locality Matching (BLM) rules for addresses on the border between two localities. To comply with the rules, Address Verification does not change an address that falls on the border of two localities in certified mode. Address Verification returns a process status code of Ix for the address and copies the input to the output elements.

For example, Address Verification returns an I4 status code for the following address because the address is located on the border of the Ramsgate Beach and Sans Souci localities:

```
138 Chuter Ave
Ramsgate Beach NSW 2217
AUS
```

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